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ICS:

Descriptors:

ENGLISH VERSION

Aerospace series — Bolt — Countersunk head, Short thread — Inch series

**Luft- und Raumfahrt — Bolzen — Senkkopf,
 Kurzgewinde — Zoll-Reihe**

**Série aéronautique — Vis — tête fraisée,
 filetage court — Série pouce**

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Mechanical Domain

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Foreword

This standard was reviewed by the Domain Technical Coordinator of ASD-STAN's Mechanical Domain.

After inquiries and votes carried out in accordance with the rules of ASD-STAN defined in ASD-STAN's General Process Manual, this standard has received approval for Publication.

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Introduction

This document is published as edition P11. Former editions P1 to P10 and drafts may exist for Airbus development only but without any ASD-STAN official publication. In consequence configuration management discrepancies with these unofficial documents are under Airbus responsibility.

1 Scope

This product standard specifies the dimensions, tolerances and the requirements of a countersunk head external threaded fastener for aerospace applications.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 2424, *Aerospace series — Marking of aerospace products*

EN 4473, *Aerospace series — Aluminium pigmented coatings for fasteners — Technical specification*

EN 6116, *Aerospace series — Fasteners — External short thread, Inch series — Technical specification*¹

EN 6117, *Aerospace series — Specification for lubrication of fasteners with cetyl alcohol*¹

EN 6118, *Aerospace series — Process specification — Aluminium base protection for fasteners*¹

ISO 8080, *Aerospace — Anodic treatment of titanium and titanium alloys — Sulphuric acid process*

AMS2700, *Passivation of Corrosion Resistant Steels*²

AMS2759/1, *Heat Treatment of Carbon and Low-Alloy Steel Parts Minimum Tensile Strength Below 220 ksi (1 517 MPa)*²

AMS4928, *Titanium Alloys Bars, Wire, Forgings, Rings, and Drawn Shapes 6Al - 4V Annealed*²

AMS4967, *Titanium Alloy, Bars, Wire, Forgings, and Rings 6.0Al - 4.0V Annealed, Heat Treatable*²

AMS5662, *Nickel Alloy, Corrosion and Heat-Resistant, Bars, Forgings, and Rings 52.5Ni - 19Cr - 3.0Mo - 5.1Cb (Nb) - 0.90Ti - 0.50Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1 775 °F (968 °C) Solution Heat Treated, Precipitation-Hardenable*²

AMS5962, *Nickel Alloy, Corrosion and Heat-Resistant, Round Bars and Wire 52.5Ni - 19Cr - 3.0Mo - 5.1Cb - 0.90Ti - 0.50Al - 19Fe Consumable Electrode or Vacuum Induction Melted 1 775 °F (968 °C) Solution Treated and Work Strengthened Precipitation Hardenable*²

AMS6322, *Steel Bars, Forgings, and Rings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740)*²

AMS6325, *Steel, Bars, and Forgings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740) Heat Treated, 105 ksi (724 MPa) Tensile Strength*²

AMS6327, *Steel Bars and Forgings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740) Heat Treated, 125 ksi (862 MPa) Tensile Strength*²

¹ Published as ASD-STAN Prestandard at the date of publication of this standard (www.asd-stan.org/).

² Published by: Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001, USA.

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AMS6349, *Steel Bars 0.95Cr - 0.20Mo (0.38 to 0.43C) (SAE 4140), Normalized*²

AMS6382, *Steel, Bars, Forgings, and Rings 0.95Cr - 0.20Mo (0.38 - 0.43C) (SAE 4140) Annealed*²

AMS6415, *Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)*²

AMS6484, *Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340) Normalized and Tempered*²

AMSQQ-P-416, *Plating, Cadmium (Electrodeposited)*²

AS8879, *Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter*²

ASME B46.1, *Surface Texture (Surface Roughness, Waviness, and Lay)*³

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at <http://www.iso.org/obp>

— IEC Electropedia: available at <http://www.electropedia.org/>

4 Requirements

4.1 Configuration, dimensions and tolerances

Configuration dimensions and tolerances shall be in accordance with Figure 1, Table 3 and Table 4.

Dimensions and tolerances for oversize fasteners shall be in accordance with Table 5, Table 6, and Table 7.

Dimensions shall apply after coating but before lubrication.

Break sharp edges with 0,25 mm max radius.

4.2 Mass

The calculation of the mass of an external threaded fastener shall be provided according to indications hereafter:

Add the mass of the head and threaded part (invariable mass) to the mass of the smooth part (variable mass).

Total mass of the head and threaded part:

1st mass column of Table 3.

3 Published by: American Society of Mechanical Engineers (ASME), Two Park Avenue, New York, NY 10016-5990, USA.

Mass of the smooth part:

Multiply the value of the 2nd mass column of Table 3 (value according to the diameter code number) by the length code number of the external threaded fastener.

EXAMPLES:

EN6114-4-8

Invariable mass	2,40 g
Variable mass	$0,39 \text{ g} \times 8 = 3,12 \text{ g}$
Head mass to be deducted	<u>-0,40 g</u>
Total mass	5,12 g

EN6114V4-8, EN6114T4-8, EN6114K4-8 or EN6114B4-8

Invariable mass	1,36 g
Variable mass	$0,22 \text{ g} \times 8 = 1,76 \text{ g}$
Head mass to be deducted	<u>-0,27 g</u>
Total mass	2,85 g

EN6114L4-8, EN6114P4-8 or EN6114M4-8

Invariable mass	2,53 g
Variable mass	$0,42 \text{ g} \times 8 = 3,36 \text{ g}$
Head mass to be deducted	<u>-0,49 g</u>
Total mass	5,40 g

EN6114L4X8

Invariable mass	2,55 g
Variable mass	$0,45 \text{ g} \times 8 = 3,60 \text{ g}$
Head mass to be deducted	<u>-0,49 g</u>
Total mass	5,66 g

4.3 Materials, finishes, lubrication and identifications

Table 1 — Materials, finishes, lubrication and identifications

Code	Material	Finish	Lubrication	Recess		Dia. range	Fastener identification
				Form	Code		
T	Titanium alloy 6Al-4V according to AMS4928 or AMS4967 or equivalent R_c min. = 650 MPa	Sulfuric-acid anodizing according to ISO 8080	Cetyl alcohol according to EN 6117	Hexagonal	-	2 to 16	None
V		IVD according to EN 6118					
K		Resin based Aluminium coating according to EN 4473 Type II ^a		5 Lobe high torque recess	E	2 to 4	A white paint identification at thread end
B		Sulfuric-acid anodizing according to ISO 8080 + Resin based Aluminium coating according to EN 4473 Type II ^a					
				Hexagonal	-	2 to 4	None
						5 to 16	

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Code	Material	Finish on threads	Lubrication	Recess		Dia. range	Fastener identification
				Form	Code		
-	Alloy steel 4340 (AMS6415 or AMS6484) or 4140 (AMS6382 or AMS6349) or 8740 (AMS6322 or AMS6325 or AMS6327) or equivalent R_c min. = 740 MPa R = 1 240 to 1 380 MPa (AMS2759/1) Hardness HRC 40 to HRC 44	Cadmium plating according to AMSQQP416, Type II, class 2			-	3 to 22	A green paint identification at thread end
L	Nickel based alloy ^b (AMS5962 or AMS5662 + cold working) R min.= 1 510 MPa	Resin based Aluminium according to EN 4473 Type II ^a			-	3 to 16	A blue paint identification at thread end
M		IVD according to EN 6118					A black paint identification at thread end
P		Passivation according to AMS2700					A yellow paint identification at thread end

^a Parts produced before 1 January 2017 may be produced with EN 4473 Type I.
^b e. g. Inconel 718. Inconel® is a trade name of a product supplied by Special Metals Corporation. This information is given for the convenience of users of this standard and does not constitute an endorsement by ASD-STAN nor CEN of the product named. Equivalent products may be used if they can be shown to lead to the same results.

4.4 Mechanical characteristics

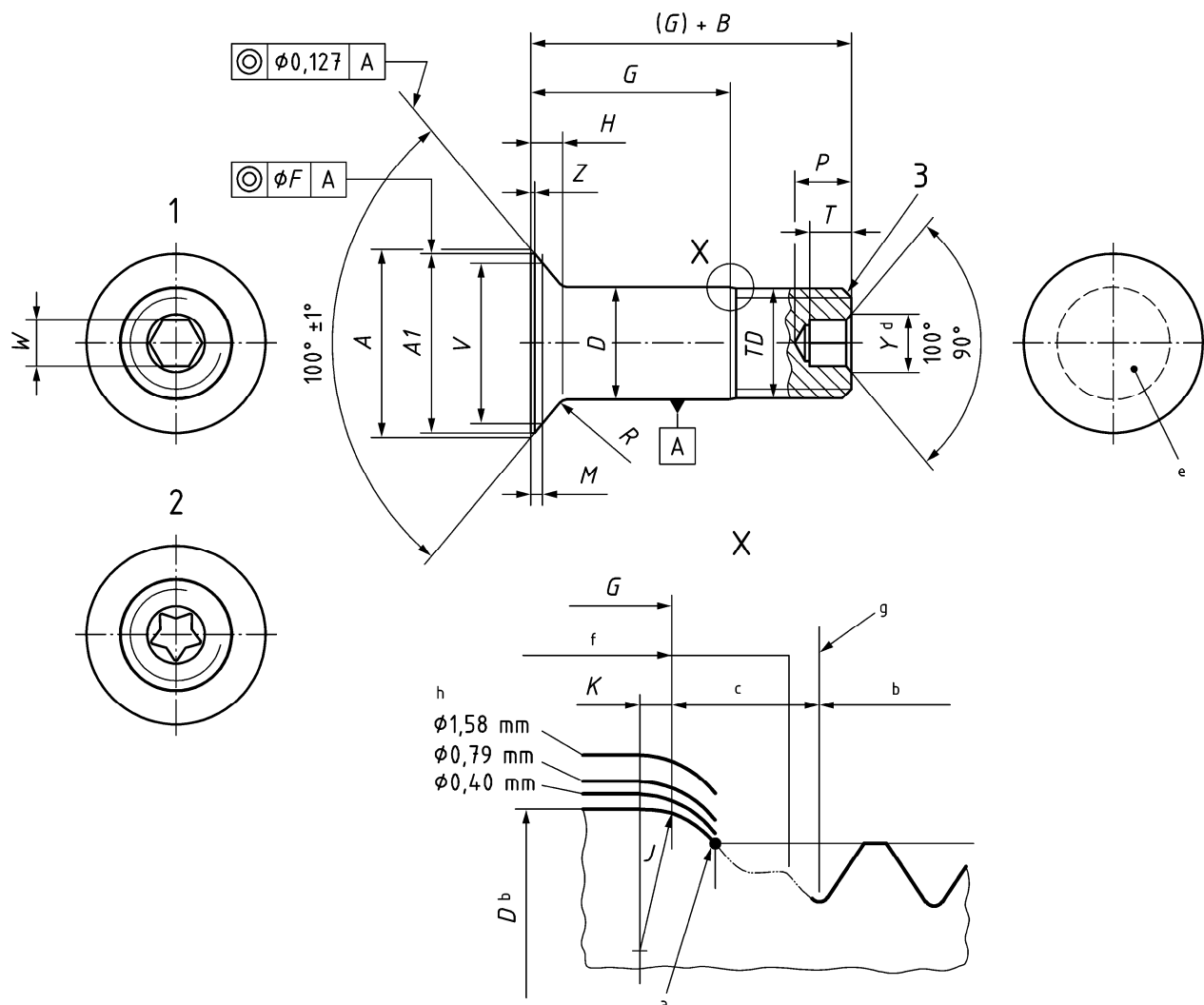
Table 2 — Mechanical characteristics

Dia. code No.	Min. double shear strength (N)			Min. tensile strength (N)			Max. fatigue load (N)		
	Steel alloy	Titanium alloy	Nickel- based alloy	Steel alloy	Titanium alloy	Nickel- based alloy	Steel alloy	Titanium alloy	Nickel- based alloy
2	–	17 760	–	–	7 310	–	–	2 340	–
3	27 250	23 900	31 500	11 500	10 700	14 000	4 050	4 000	4 940
3A	35 400	32 000	–	16 300	14 450	–	5 700	5 050	–
4	47 150	41 330	54 800	22 250	20 000	27 090	7 800	7 000	9 550
5	73 850	64 880	85 100	33 350	30 450	40 600	11 700	10 650	14 310
6	106 300	93 320	123 000	48 950	45 350	59 600	17 150	15 900	21 010
7	144 550	127 100	167 300	63 600	58 250	77 440	22 250	20 400	27 300
8	188 600	165 760	218 300	88 100	80 000	107 280	30 850	28 000	37 820
9	238 850	209 950	276 230	109 450	100 000	126 320	38 250	35 050	44 210
10	294 900	259 330	341 300	137 900	129 900	167 920	48 250	45 350	59 210
12	424 350	373 200	491 130	213 500	186 340	259 980	74 750	65 700	91 030
14	573 800	493 990	664 100	289 150	250 290	326 940	101 200	88 260	114 420
16	749 500	625 970	867 450	378 100	326 340	411 460	132 350	115 070	144 000
18	951 900	–	–	487 056	–	–	169 024	–	–
20	1 178 800	–	–	632 100	–	–	214 850	–	–
22	1 423 400	–	–	776 200	–	–	263 800	–	–
NOTE Minimum fatigue loads are equal to 10 % of maximum loads.									
NOTE Mechanical characteristic values also apply to oversize fasteners.									

4.5 General characteristics

The surface condition shall be according to ASME B46.1.

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Key

- 1 Hexagonal recess code “-” (see Table 1)
- 2 5 Lobe high torque recess code “E” (see Table 1)
- 3 Chamfer S

NOTE The maximum thread runout and functionality for 1st oversize is increased by 0,25 mm, for 2nd oversize by 0,5 mm, and for 3rd oversize by 1 mm.

- a For nominal diameter: The diameter measured at point I shall be less than or equal to the maximum diameter *TD* for nominal diameter. For oversizes diameters: Maximum diameter at point I shall be incremented by 0,40 mm, 0,79 mm and 1,58 mm for respectively 1st, 2nd, and 3rd oversize. The *TD* diameter stays the same as for nominal diameter.
- b For B coded fasteners only threads shall be coated with aluminium pigmented resin-based coating.
- c For B coded fasteners overspray is permissible in this area.
- d Valid for hexagonal recess.
- e Marking
- f Thread functionality 1,0 pitch
- g Thread runout max. 1,5 pitch
- h Oversizes

Figure 1 — Configuration, dimensions and tolerances

Table 3 — Dimensions, tolerances and mass

Dimensions in millimetres

Dia- meter code No.	Nominal shank diameter	Thread ^a UNJF-3A modified (inch)	A max.	A1 min.	B ref.	D		TD	F max.	H ^c ref.	R
						T, B or P code	Other code				
2	3,97	0.1640-32 ^b	7,18	6,60	7,11	4,153 4,140	4,153 4,128	4,051 3,988	0,10	1,24	0,64 0,38
3	4,76	0.1900-32	8,32	7,44	7,37	4,813 4,800	4,813 4,788	4,673 4,597	0,13	1,42	0,76 0,51
3A	5,56	0.2160-28	9,57	8,69	7,75	5,542 5,529	5,542 5,517	5,334 5,258	0,15	1,65	
4	6,35	0.2500-28	10,88	10,01	8,13	6,337 6,324	6,337 6,312	6,197 6,121		1,88	
5	7,94	0.3125-24	13,62	12,73	9,65	7,925 7,911	7,925 7,899	7,772 7,670	0,18	2,34	1,02 0,76
6	9,53	0.3750-24	16,29	15,42	10,67	9,512 9,500	9,512 9,487	9,347 9,245	0,20	2,79	
7	11,11	0.4375-20	18,86	17,55	12,32	11,099 11,087	11,099 11,074	10,947 10,820	0,23	3,20	1,27 1,02
8	12,70	0.5000-20	21,39	20,09	13,33	12,687 12,674	12,687 12,662	12,522 12,395	0,25	3,61	
9	14,29	0.5625-18	23,62	22,38	15,24	14,262 14,249	14,262 14,237	14,097 13,970		3,88	
10	15,88	0.6250-18	26,52	25,27	16,26	15,849 15,836	15,849 15,824	15,697 15,545		4,42	
12	19,05	0.7500-16	33,02	31,78	22,73	19,024 19,011	19,024 18,999	18,872 18,719	0,30	5,82	
14	22,23	0.8750-14	38,33	37,11	25,40	22,200 22,187	22,200 22,174	22,047 21,869	0,36	6,68	
16	25,40	1.0000-12	43,69	42,44	29,46	25,375 25,362	25,375 25,349	25,222 25,044		7,57	
18	28,58	1.1250-12	49,15	47,93	33,15	28,549 28,537	28,549 28,524	28,372 28,194	0,38	8,64	1,52 1,27
20	31,75	1.2500-12	55,00	54,00	36,54	31,725 31,712	31,725 31,700	31,532 31,232		9,81	
22	34,93	1.3750-12	61,00	60,00	39,64	34,900 34,887	34,900 34,875	34,706 34,382		10,96	
^a Thread according to AS8879 except diameter <i>TD</i> . ^b Thread UNJC-3A. ^c Height <i>H</i> is dimensioned based on max. diameter <i>D</i> .											

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Table 3 — Dimensions, tolerances and mass *(continued)*

Dimensions in millimetres

Dia. code No.	Nom. shank dia.	Thread ^a UNJF-3A modified (inch)	Z max.	S ^e ref.	Hexagonal recess				5 Lobe high torque recess		M	V
					P max.	W	T	Y	T min.	P max.		
2	3,97	0.1640-32 ^b	0,25	0,79	3,43	2,05 2,01	2,54 2,04	2,64 2,39	1,73	3,03	0,848 0,757	5,151 5,146
3	4,76	0.1900-32	0,38					3,02 2,64	1,77	3,10	0,749 0,668	6,502 6,497
3A	5,56	0.2160-28						1,80	3,30	0,820 0,744	7,574 7,569	
4	6,35	0.2500-28			3,78	2,46 2,41	2,79 2,29	3,61 3,10	1,97	3,50	0,576 0,495	9,479 9,474
5	7,94	0.3125-24	0,56	1,19	4,14	3,29 3,23	2,99 2,49	4,57 4,06	–	–	0,594 0,503	12,169 12,164
6	9,53	0.3750-24			5,26	4,11 4,02	3,68 3,18	5,51 5,00			0,749 0,657	14,473 14,467
7	11,11	0.4375-20			5,97	4,90 4,81	4,36 3,86	6,43 5,92			0,881 0,779	16,718 16,713
8	12,70	0.5000-20			6,88	5,69 5,61	5,05 4,55	7,34 6,83			1,280 1,178	18,288 18,283
9	14,29	0.5625-18	1,59	8,08	6,49 6,40	5,97 5,47	8,28 7,77	1,353 1,232			20,350 20,345	
10	15,88	0.6250-18						1,602 1,496			22,611 22,606	
12	19,05	0.7500-16						1,971 1,818			28,254 28,249	
14	22,23	0.8750-14						1,762 1,579			34,137 34,132	
16	25,40	1.0000-12	1,98	15,80	12,95 12,83	11,68 11,18	15,70 15,19	1,567 1,361			39,959 39,954	
18	28,58	1.1250-12						1,422 1,206			45,786 45,781	
20	31,75	1.2500-12						2,10			19,75	16,13 16,00
22	34,93	1.3750-12		3,21	21,95	17,73 17,60	15,66 15,16	21,14 20,94				

^a Thread according to AS8879 except diameter *TD*.^b Thread UNJC-3A.^c See Figure 1.^d Height *H* is dimensioned based on max. diameter *D*.^e 37° to 45°.

Table 3 — Dimensions, tolerances and mass (continued)

Dimensions in millimetres

Dia. code No.	Nom. shank dia.	Thread ^a UNJF-3A modified (inch)	Detail A		Mass ref. (g)								
			<i>J</i>	<i>K</i> max.	Head and thread			Smooth part			Head mass to be deducted		
					Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy
2	3,97	0.1640-32 ^b	2,16 1,65	0,33	–	0,44	–	–	0,10	–	–	0,07	–
3	4,76	0.1900-32	2,67 1,78	0,41	1,15	0,65	1,22	0,23	0,13	0,24	0,20	0,12	0,21
3A	5,56	0.2160-28	2,92 2,41	0,48	–	0,94	–	–	0,17	–	–	0,18	–
4	6,35	0.2500-28	3,68 3,18	0,53	2,40	1,36	2,53	0,39	0,22	0,42	0,40	0,27	0,49
5	7,94	0.3125-24	4,45 3,94	0,66	4,60	2,61	4,83	0,61	0,35	0,65	0,90	0,52	0,95
6	9,53	0.3750-24	5,97 5,46	0,76	7,64	4,32	8,02	0,88	0,50	0,93	1,55	0,88	1,63
7	11,11	0.4375-20	7,75 7,24	0,89	11,73	6,64	12,32	1,20	0,69	1,27	2,42	1,37	2,55
8	12,70	0.5000-20	9,14 8,63	0,99	17,01	9,62	17,86	1,57	0,89	1,65	3,57	2,02	3,75
9	14,29	0.5625-18	9,78 9,27		23,78	13,45	24,96	1,99	1,13	2,09	4,86	2,75	5,10
10	15,88	0.6250-18	10,03 9,40	1,04	33,22	18,80	34,86	2,45	1,39	2,58	6,83	3,87	7,17
12	19,05	0.7500-16	10,29 9,53	1,12	67,27	38,06	70,62	3,54	2,00	3,71	12,95	7,33	13,60
14	22,23	0.8750-14	10,41 9,65	1,14	100,53	58,09	108,17	4,82	2,73	5,06	20,24	11,46	21,26
16	25,40	1.0000-12	11,18 10,41		145,41	84,36	156,45	6,29	3,56	6,61	29,97	16,96	31,47
18	28,58	1.1250-12	12,18 11,18	1,19	214,62	121,42	225,31	7,96	4,50	8,35	43,31	24,50	45,46
20	31,75	1.2500-12	12,95 12,18		297,05	168,07	311,85	9,83	5,56	10,32	60,72	34,35	63,74
22	34,93	1.3750-12		1,22	400,56	226,63	420,52	11,89	6,73	12,48	81,64	46,19	85,71
^a Thread according to AS8879 except diameter <i>TD</i> .													
^b Thread UNJC-3A.													

Table 4 — Dimensions and tolerances

Dimensions in millimetres

Length code No. ^a	G ±0,13	Length (G + B ref.) ± 0,25							
		2	3	3A	4	5	6	7	8
1	1,59	8,70	8,95	9,34	9,72	11,24	–	–	–
2	3,18	10,29	10,54	10,93	11,31	12,83	13,85	–	–
3	4,76	11,87	12,12	12,51	12,89	14,41	15,43	17,08	–
4	6,35	13,46	13,71	14,10	14,48	16,00	17,02	18,67	19,68
5	7,94	15,05	15,30	15,69	16,07	17,59	18,61	20,26	21,27
6	9,52	16,63	16,88	17,27	17,65	19,17	20,19	21,84	22,85
7	11,11	18,22	18,47	18,86	19,24	20,76	21,78	23,43	24,44
8	12,70	19,81	20,06	20,45	20,83	22,35	23,37	25,02	26,03
9	14,29	21,40	21,65	22,04	22,42	23,94	24,96	26,61	27,62
10	15,88	22,99	23,24	23,63	24,01	25,53	26,55	28,20	29,21
11	17,46	24,57	24,82	25,21	25,59	27,11	28,13	29,78	30,79
12	19,05	26,16	26,41	26,80	27,18	28,70	29,72	31,37	32,38
13	20,64	27,75	28,00	28,39	28,77	30,29	31,31	32,96	33,97
14	22,22	29,33	29,58	29,97	30,35	31,87	32,89	34,54	35,55
15	23,81	30,92	31,17	31,56	31,94	33,46	34,48	36,13	37,14
16	25,40	32,51	32,76	33,15	33,53	35,05	36,07	37,72	38,73
17	26,99	34,10	34,35	34,74	35,12	36,64	37,66	39,31	40,32
18	28,58	35,69	35,94	36,33	36,71	38,23	39,25	40,90	41,91
19	30,16	37,27	37,52	37,91	38,29	39,81	40,83	42,48	43,49
20	31,75	38,86	39,11	39,50	39,88	41,40	42,42	44,07	45,08
21	33,34	40,45	40,70	41,09	41,47	42,99	44,01	45,66	46,67
22	34,92	42,03	42,28	42,67	43,05	44,57	45,59	47,24	48,25
23	36,51	43,62	43,87	44,26	44,64	46,16	47,18	48,83	49,84
24	38,10	45,21	45,46	45,85	46,23	47,75	48,77	50,42	51,43
25	39,69	46,80	47,05	47,44	47,82	49,34	50,36	52,01	53,02
26	41,28	48,39	48,64	49,03	49,41	50,93	51,95	53,60	54,61
27	42,86	49,97	50,22	50,61	50,99	52,51	53,53	55,18	56,19
28	44,45	51,56	51,81	52,20	52,58	54,10	55,12	56,77	57,78
29	46,04	53,15	53,40	53,79	54,17	55,69	56,71	58,36	59,37
30	47,62	54,73	54,98	55,37	55,75	57,27	58,29	59,94	60,95
31	49,21	56,32	56,57	56,96	57,34	58,86	59,88	61,53	62,54
32	50,80	57,91	58,16	58,55	58,93	60,45	61,47	63,12	64,13

Length code No. ^a	<i>G</i> ±0,13	Length (<i>G</i> + <i>B</i> ref.) ± 0,25							
		2	3	3A	4	5	6	7	8
34	53,98	61,09	61,34	61,73	62,11	63,63	64,65	66,30	67,31
36	57,15	64,26	64,51	64,90	65,28	66,80	67,82	69,47	70,48
38	60,32	67,43	67,68	68,07	68,45	69,97	70,99	72,64	73,65
40	63,50	70,61	70,86	71,25	71,63	73,15	74,17	75,82	76,83
42	66,68	73,79	74,04	74,43	74,81	76,33	77,35	79,00	80,01
44	69,85	76,96	77,21	77,60	77,98	79,50	80,52	82,17	83,18
46	73,02	80,13	80,38	80,77	81,15	82,67	83,69	85,34	86,35
48	76,20	83,31	83,56	83,95	84,33	85,85	86,87	88,52	89,53
50	79,38	86,49	86,74	87,13	87,51	89,03	90,05	91,70	92,71
52	82,55	89,66	89,91	90,30	90,68	92,20	93,22	94,87	95,88
54	85,72	92,83	93,08	93,47	93,85	95,37	96,39	98,04	99,05
56	88,90	96,01	96,26	96,65	97,03	98,55	99,57	101,22	102,23
58	92,08	99,19	99,44	99,83	100,21	101,73	102,75	104,40	105,41
60	95,25	102,36	102,61	103,00	103,38	104,90	105,92	107,57	108,58
62	98,43	105,54	105,80	106,18	106,56	108,08	109,10	110,75	111,76
64	101,60	108,71	108,97	109,35	109,73	111,25	112,27	113,92	114,93
66	104,78	111,89	112,15	112,53	112,91	114,43	115,45	117,10	118,11
68	107,95	115,06	115,32	115,70	116,08	117,60	118,62	120,27	121,28
70	111,13	118,24	118,50	118,88	119,26	120,78	121,80	123,45	124,46
72	114,30	121,41	121,67	122,05	122,43	123,95	124,97	126,62	127,63
74	117,48	124,59	124,85	125,23	125,61	127,13	128,15	129,80	130,81
76	120,65	127,76	128,02	128,40	128,78	130,30	131,32	132,97	133,98
78	123,83	130,94	131,20	131,58	131,96	133,48	134,50	136,15	137,16
80	127,00	134,11	134,37	134,75	135,13	136,65	137,67	139,32	140,33
^a Intermediate grip lengths may be purchased in 1,59 mm increment if necessary.									

Table 4 — Dimensions and tolerances (continued)

Dimensions in millimetres

Length code No. ^a	G ±0,13	Length (G + B ref.) ± 0,25							
		9	10	12	14	16	18	20	22
1	1,59	–	–	–	–	–	–	–	–
2	3,18	–	–	–	–	–	–	–	–
3	4,76	–	–	–	–	–	–	–	–
4	6,35	–	–	–	–	–	–	–	–
5	7,94	23,18	–	–	–	–	–	–	–
6	9,52	24,76	25,77	–	–	–	–	–	–
7	11,11	26,35	27,36	33,84	–	–	–	–	–
8	12,70	27,94	28,95	35,43	38,10	–	–	–	–
9	14,29	29,53	30,54	37,02	39,69	43,75	–	–	–
10	15,88	31,12	32,13	38,61	41,28	45,34	49,03	–	–
11	17,46	32,70	33,71	40,19	42,86	46,92	50,61	54,00	–
12	19,05	34,29	35,30	41,78	44,45	48,51	52,20	55,59	58,69
13	20,64	35,88	36,89	43,37	46,04	50,10	53,79	57,18	60,28
14	22,22	37,46	38,47	44,95	47,62	51,68	55,37	58,76	61,86
15	23,81	39,05	40,06	46,54	49,21	53,27	56,96	60,35	63,45
16	25,40	40,64	41,65	48,13	50,80	54,86	58,55	61,94	65,04
17	26,99	42,23	43,24	49,72	52,39	56,45	60,14	63,53	66,63
18	28,58	43,82	44,83	51,31	53,98	58,04	61,73	65,12	68,22
19	30,16	45,40	46,41	52,89	55,56	59,62	63,31	66,70	69,80
20	31,75	46,99	48,00	54,48	57,15	61,21	64,90	68,29	71,39
21	33,34	48,58	49,59	56,07	58,74	62,80	66,49	69,88	72,98
22	34,92	50,16	51,17	57,65	60,32	64,38	68,07	71,46	74,56
23	36,51	51,75	52,76	59,24	61,91	65,97	69,66	73,05	76,15
24	38,10	53,34	54,35	60,83	63,50	67,56	71,25	74,64	77,74
25	39,69	54,93	55,94	62,42	65,09	69,15	72,84	76,23	79,33
26	41,28	56,52	57,53	64,01	66,68	70,74	74,43	77,82	80,92
27	42,86	58,10	59,11	65,59	68,26	72,32	76,01	79,40	82,50
28	44,45	59,69	60,70	67,18	69,85	73,91	77,60	80,99	84,09
29	46,04	61,28	62,29	68,77	71,44	75,50	79,19	82,58	85,68
30	47,62	62,86	63,87	70,35	73,02	77,08	80,77	84,16	87,26
31	49,21	64,45	65,46	71,94	74,61	78,67	82,36	85,75	88,85
32	50,80	66,04	67,05	73,53	76,20	80,26	83,95	87,34	90,44

Length code No. ^a	G ±0,13	Length (G + B ref.) ± 0,25							
		9	10	12	14	16	18	20	22
34	53,98	69,22	70,23	76,71	79,38	83,44	87,13	90,52	93,62
36	57,15	72,39	73,40	79,88	82,55	86,61	90,30	93,69	96,79
38	60,32	75,56	76,57	83,05	85,72	89,78	93,47	96,86	99,96
40	63,50	78,74	79,75	86,23	88,90	92,96	96,65	100,04	103,14
42	66,68	81,92	82,93	89,41	92,08	96,14	99,83	103,22	106,32
44	69,85	85,09	86,10	92,58	95,25	99,31	103,00	106,39	109,49
46	73,02	88,26	89,27	95,75	98,42	102,48	106,17	109,56	112,66
48	76,20	91,44	92,45	98,93	101,60	105,66	109,35	112,75	115,84
50	79,38	94,62	95,63	102,11	104,78	108,84	112,53	115,92	119,02
52	82,55	97,79	98,80	105,28	107,95	112,01	115,70	119,09	122,19
54	85,72	100,96	101,97	108,45	111,12	115,18	118,87	122,26	125,36
56	88,90	104,14	105,15	111,63	114,30	118,36	122,05	125,44	128,54
58	92,08	107,32	108,33	114,81	117,48	121,54	125,23	128,62	129,72
60	95,25	110,49	111,50	117,98	120,65	124,71	128,40	131,79	134,89
62	98,43	113,67	114,69	121,16	123,83	127,89	131,58	134,97	138,07
64	101,60	116,84	117,86	124,33	127,00	131,06	134,75	138,62	141,24
66	104,78	120,02	121,04	127,51	130,18	134,24	137,93	141,32	144,42
68	107,95	123,19	124,21	130,68	133,35	137,41	141,10	144,49	147,59
70	111,13	126,37	127,39	133,86	136,53	140,59	144,28	147,67	150,77
72	114,30	129,54	130,56	137,03	139,70	143,76	147,45	150,84	153,94
74	117,48	132,72	133,74	140,21	142,88	146,94	150,63	154,02	157,12
76	120,65	135,89	136,91	143,38	146,05	150,11	153,80	157,19	160,29
78	123,83	139,07	140,09	146,56	149,23	153,29	156,98	160,37	163,47
80	127,00	142,24	143,26	149,73	152,40	156,46	160,15	163,54	166,64
^a Intermediate grip lengths may be purchased in 1,59 mm increment if necessary.									

prEN 6114:2019 (E)**Table 5 — Oversize dimensions 1st Oversize**

Dimensions in millimetres

Thread ^a UNJF-3A modified (inch)	Oversize shank ^b						Mass Ref. (g)								
	Dia. code No.	Nom. dia.	B ref.	D		H ref.	Head and thread			Smooth part			Head mass to be deducted		
				T, B or P code	Other code		Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy
0.1900-32	3X	5,16	7,62	5,146 5,134	5,146 5,121	1,32	1,13	0,65	1,22	0,24	0,14	0,26	0,20	0,12	0,22
0.2160-28	3AX	5,95	8,00	5,939 5,927	5,939 5,914	1,50	–	0,94	–	–	0,18	–	–	0,18	–
0.2500-28	4X	6,75	8,38	6,734 6,722	6,734 6,708	1,73	2,37	1,37	2,55	0,41	0,23	0,45	0,46	0,27	0,49
0.3125-24	5X	8,33	9,91	8,321 8,309	8,321 8,296	2,21	4,52	2,61	4,86	0,64	0,37	0,69	0,90	0,52	0,97
0.3750-24	6X	9,92	10,92	9,909 9,897	9,909 9,883	2,67	7,49	4,32	8,06	0,92	0,53	0,99	1,54	0,89	1,66
0.4375-20	7X	11,51	12,57	11,496 11,484	11,496 11,471	3,07	11,61	6,70	12,49	1,24	0,71	1,34	2,41	1,39	2,59
0.5000-20	8X	13,10	13,58	13,084 13,072	13,084 13,058	3,48	16,35	9,43	17,59	1,61	0,93	1,73	3,55	2,05	3,82
0.5625-18	9X	14,68	15,49	14,658 14,646	14,658 14,633	3,73	23,35	13,47	25,12	2,02	1,17	2,18	4,83	2,78	5,20
0.6250-18	10X	16,27	16,51	16,246 16,234	16,246 16,220	4,27	32,63	18,83	35,11	2,50	1,45	2,70	6,79	3,92	7,31
0.7500-16	12X	19,45	22,98	19,421 19,409	19,421 19,396	5,64	66,01	37,60	71,02	3,59	2,05	3,87	12,89	7,34	13,87
0.8750-14	14X	22,62	25,65	22,596 22,584	22,596 22,571	6,53	100,53	57,26	108,17	4,88	2,78	5,24	20,14	11,47	21,67
1.0000-12	16X	25,80	29,72	25,771 25,759	25,771 25,746	7,42	145,41	82,83	156,45	6,36	3,63	6,85	29,83	16,99	32,10
1.1250-12	18X	28,97	33,40	28,945 28,933	28,945 28,920	8,48	214,62	121,42	225,31	8,02	4,56	8,47	43,31	24,50	45,46
1.2500-12	20X	32,15	36,79	32,121 32,109	32,121 32,097	9,64	297,05	168,07	311,85	9,95	5,63	10,45	60,72	34,35	63,74
1.3750-12	22X	35,32	39,89	35,296 35,284	35,296 35,272	10,80	400,56	226,63	420,52	12,03	6,81	12,62	81,64	46,19	85,71
^a Thread according to AS8879 except diameter <i>TD</i> .															
^b For diameter code 3X, shank diameter is incremented by 0,333 mm.															

Table 6 — Oversize dimensions 2nd Oversize

Dimensions in millimetres

Thread ^a UNJF-3A modified (inch)	Oversize shank						Mass Ref. (g)								
	Dia. code No.	Nom. dia.	B ref.	D		H ref.	Head and thread			Smooth part			Head mass to be deducted		
				T, B or P code	Other code		Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy
0.1900-32	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
0.2160-28	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
0.2500-28	4Y	7,14	8,38	7,130 7,118	7,130 7,104	1,52	2,37	1,37	2,55	0,44	0,25	0,47	0,46	0,27	0,49
0.3125-24	5Y	8,73	9,91	8,717 8,705	8,717 8,692	2,01	4,52	2,61	4,86	0,67	0,39	0,73	0,90	0,52	0,97
0.3750-24	6Y	10,32	10,92	10,305 10,293	10,305 10,279	2,46	7,49	4,32	8,06	0,95	0,55	1,03	1,54	0,89	1,66
0.4375-20	7Y	11,91	12,57	11,892 11,880	11,892 11,867	2,90	11,61	6,70	12,49	1,29	0,74	1,38	2,41	1,39	2,59
0.5000-20	8Y	13,49	13,58	13,480 13,468	13,480 13,454	3,25	16,35	9,43	17,59	1,66	0,96	1,79	3,55	2,05	3,82
0.5625-18	9Y	15,08	15,49	15,055 15,043	15,055 15,029	3,56	23,35	13,47	25,12	2,08	1,20	2,24	4,83	2,78	5,20
0.6250-18	10Y	16,67	16,51	16,642 16,630	16,642 16,617	4,09	32,63	18,83	35,11	2,56	1,48	2,76	6,79	3,92	7,31
0.7500-16	12Y	19,84	22,98	19,817 19,805	19,817 19,792	5,46	66,01	37,60	71,02	3,67	2,09	3,95	12,89	7,34	13,87
0.8750-14	14Y	23,02	25,65	22,992 22,980	22,992 22,967	6,10	100,53	57,26	108,17	4,96	2,83	5,33	20,14	11,47	21,67
1.0000-12	16Y	26,19	29,72	26,167 26,155	26,167 26,142	7,24	145,41	82,83	156,45	6,46	3,68	6,95	29,83	16,99	32,10
1.1250-12	18Y	29,37	33,40	29,342 29,330	29,342 29,317	8,31	214,62	121,42	225,31	8,13	4,63	8,58	43,31	24,50	45,46
1.2500-12	20Y	32,54	36,79	32,517 32,505	32,517 32,493	9,48	297,05	168,07	311,85	10,08	5,70	10,58	60,72	34,35	63,74
1.3750-12	22Y	35,72	39,89	35,692 35,680	35,692 35,668	10,63	400,56	226,63	420,52	12,16	6,88	12,76	81,64	46,19	85,71
^a Thread according to AS-8879 except diameter <i>TD</i> .															

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Table 7 — Oversize dimensions 3rd Oversize

Dimensions in millimetres

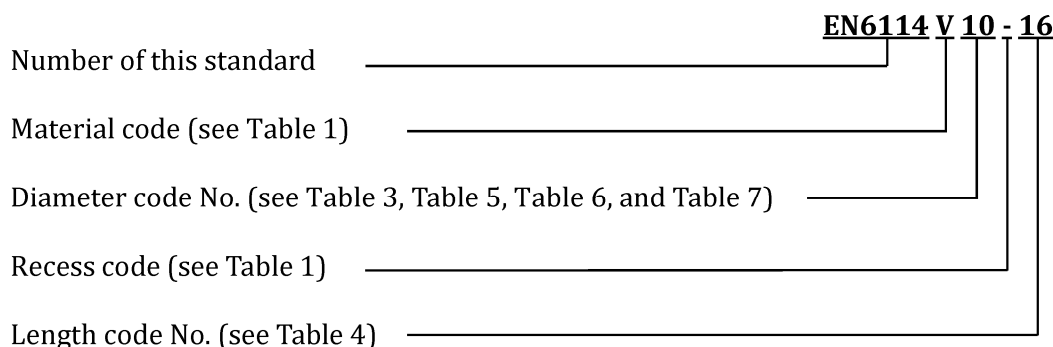
Thread ^a UNJF-3A modified (inch)	Oversize shank											Mass Ref. (g)								
	Dia. code No.	Nom. dia.	B ref.	D		H ref.	A max.	A1 min.	Z max.	M	V	Head and thread			Smooth part			Head mass to be deducted		
				T, B or P code	Other code							Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy	Steel alloy	Tita- nium alloy	Ni- based alloy
0.6250-18	10Z	17,46	16,76	17,437 17,424	17,437 17,412	5,31	30,09	28,76	0,558	3,140 3,060	22,611 22,606	32,63	18,83	35,11	2,68	1,55	2,89	6,79	3,92	7,31
0.7500-16	12Z	20,64	23,23	20,612 20,599	20,612 20,587	6,24	35,49	34,16		3,036 2,909	28,254 28,249	66,01	37,60	71,02	3,81	2,18	4,11	12,89	7,34	13,87
0.8750-14	14Z	23,81	25,90	23,787 23,775	23,787 23,762	7,18	40,90	39,57		2,837 2,716	34,137 34,132	100,53	57,26	108,17	5,13	2,93	5,52	20,14	11,47	21,67
1.0000-12	16Z	26,99	29,96	26,962 26,950	26,962 26,937	8,11	46,29	44,96		2,656 2,529	39,959 39,954	145,41	82,83	156,45	6,65	3,79	7,16	29,83	16,99	32,10

^a Thread according to AS8879 except diameter *TD*.

5 Designation

This type of standard shall be designated according to the philosophy of the following example:

EXAMPLE



If the diameter code ends with a letter (e. g. 3A, 3X, 3AX, 4X or 4Y) do not use the “-” recess code (e. g. EN6114K3A5, EN6114-3A5, EN6114B3E5, EN6114B3AXE5).

6 Marking

Parts shall be marked according to EN 2424, style B. The marking shall be recessed with maximum depth of 0,25 mm.

7 Technical specification

The technical specification shall be according to EN 6116.

8 Quality assurance

The manufacturer’s operations shall be an approved production organisation for aerospace products and shall demonstrate that it has implemented and is able to maintain a quality management system (e. g. according to EN 9100 or an equivalent aerospace accepted and established quality management system).

The qualification procedure for aerospace standard products (e. g. according to EN 9133 or an equivalent aerospace accepted and established qualification procedure) shall be used and documented according to the specified tests if not otherwise agreed between customer and supplier.

Annex A (informative)

Record of revisions

Table A.1 — Record of revisions

Edition	Clause modified	Description of modification
1 06/02		New standard.
2 07/02	§ 2 Tables 3, 4 and 5	Normative references updated. Introduction of diameter code Nos 20 and 22. Data for P max.
3 08/03	§ 3.1 Table 2	Limitation of application: Large diameter (-12 to -18) titanium fasteners Addition of mechanical characteristics for -12 Ni-based alloy fasteners.
4 07/04	§ 3.1 Tables 3 and 5	Limitation of grip length. Addition of third oversize.
5 01/08	§ 3.2 Table 1 Table 2 § 3.5 Figure 1 Table 3 Table 5	New references added. Code B added. Reference updated. Ni-based alloy mechanical properties updated for dia. code No. 9. Drawing with oversize's updated. Note 4 added for code B. Note 1 modified. Headings and masses updated. Headings updated. Mass added for oversize's dimensions.
6 11/09	2 Table 1 and 3	Normative references updated. Reference AMS QQ-P-35 replaced by AMS2770. Addition of P code.
7 08/10	Fig. 1	5 Lobe high torque recess version added
	Table 1	5 Lobe high torque recess with recess code E added for code B pins
	Table 1	Footnote updated.
	Table 1, Fig.1	Table 1 and fig. 1 updated. footnote 5 added and footnote 4B updated
	Table 3	Dimension P max. moved to column Hexagonal recess Dimensions for 5 Lobe high torque recess added Dimension T min. and P max. updated
	Table 5	Column D amended with "T, B or P code"
	Designation	EN6115B3AX5 modified to EN6115B3AXE5 Footnote a) added. Recess code added without influence of existing designation.

Edition	Clause modified	Description of modification
8 06/14	Table 1	Finish code k has been replaced by finish code F according to the ACF requirements. Diameter ranges corrected in line with Table 2.
9 11/15	Table 2	Mechanical characteristic values for diameters 14 and 16 added for Ni-based alloy material.
10 03/18	3 Table 1	Limitation of application removes as it is a design requirement For Codes K, B and L Finish amended to read EN4473 Type II Footnote 2) added.
	3.5 Table 3	Footnote for thread rolling of steel parts removed Dimension Z amended to read Z max ØY dimension for diameter 2 dimensions amended in accordance with manufacturers drawings. Dimension J for diameters 6, 8 and 18 amended in accordance with manufacturers drawings.

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Bibliography

EN 9100, *Quality Management Systems — Requirements for Aviation, Space and Defence Organizations*

EN 9133, *Aerospace series — Quality Management Systems — Qualification Procedure for Aerospace Standard Products*